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Project 2 Report

Task 1:

1. Is there any chance of deadlock in this simulation? What changes could cause deadlocks?

There is no chance of a deadlock with the implementation of the semaphores in the code. A simple change to cause a deadlock is removing the lock for the write semaphores and letting the system run. Two writers will attempt to write at the same time and be locked forever waiting for the other.

Task 2:

1. How does this task compare with Task 1? Which is easier to implement?

Task 2 in many ways was easier to implement than Task 1. Partly because most of the code was already there to be worked off of from Task 1, but also because creating the lists was easier to implement than crafting a specific matrix and searching through it.

Task 3:

1. What disadvantages do capability lists introduce given a more domain-based approach?

While capability lists are easy to understand and provide flexibility for users to define certain parameters, they struggle with large groups of users. For example, if one wished to provide 100 users access to file_3, every individual user's capability list would have to be accessed and edited to include file_3 in some way as opposed to ACL where only the one file_3 list would have to be accessed. Additionally, it is difficult to check up on the capabilities of others when concerning a file for similar reasons. If you wanted to check what every user's access to file_3 is, you would have to check each individual user's list to see if they even have access or not.

Report:

1. What is the importance of protection in a multi-user system? What could happen if all users were granted equal permission?

The importance of protection in a multi-user system is the protection of the entire system, users, and all the data the system harbors. Protections where none are needed is a surefire way to have someone messing in something they're not meant to, and thus is a major security risk. If all users were granted equal permissions, then users would have access to system code, databases, user records, and the ability to edit all of those things. The wrong person, such as someone who doesn't know what they're doing, could cause massive damage if left unchecked with those capabilities.